

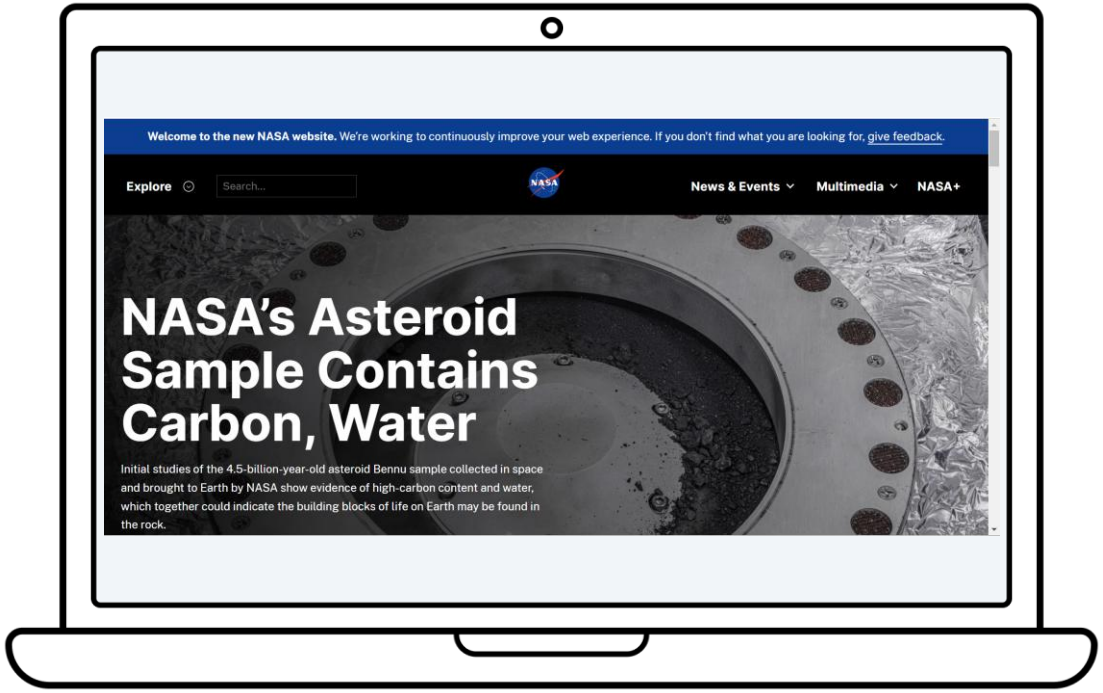
CSF3133: WEB DESIGN

ASSOC. PROF. TS. DR. WAN NURAL JAWAHIR HJ WAN YUSSOF

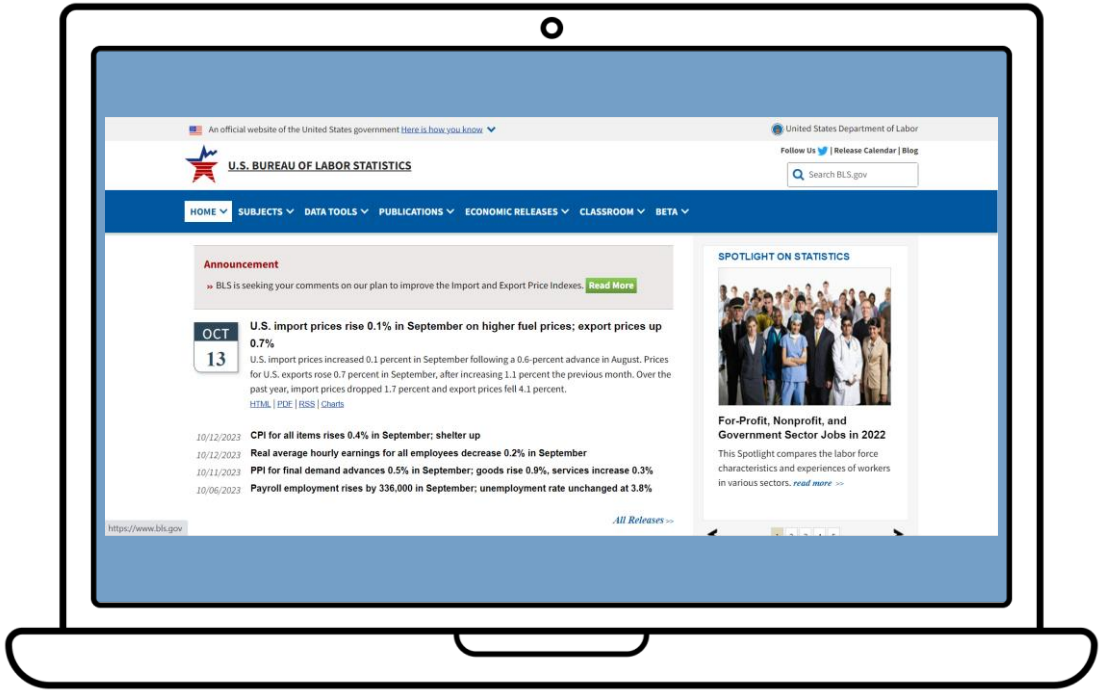
LEARNING OBJECTIVES

- Describe the most common types of website organization
- Describe principles of visual design
- Design for your target audience
- Create clear, easy-to-use navigation
- Improve the readability of the text on your web pages
- Use graphics appropriately on web pages
- Apply the concept of universal design to web pages
- Describe web page layout design techniques
- Apply best practices of web design

OVERALL DESIGN IS RELATED TO THE SITE PURPOSE



http://nasa.gov



http://bls.gov

WEBSITE ORGANIZATION



Hierarchical organization

Web pages are organized in a hierarchical tree structure with the home page at the top.



Linear organization

Web pages are organized in a linear sequence that users follow.

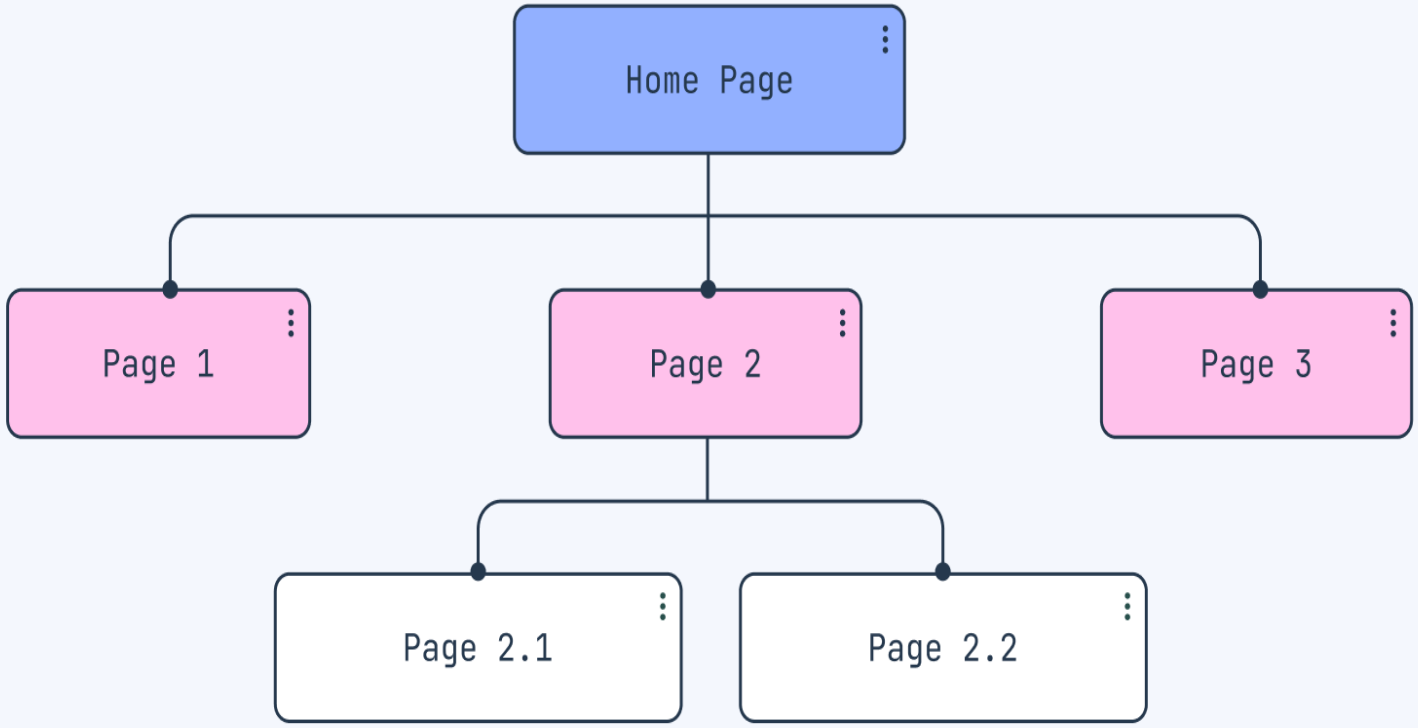


Random organization

Web pages are connected randomly with hyperlinks, like a web.

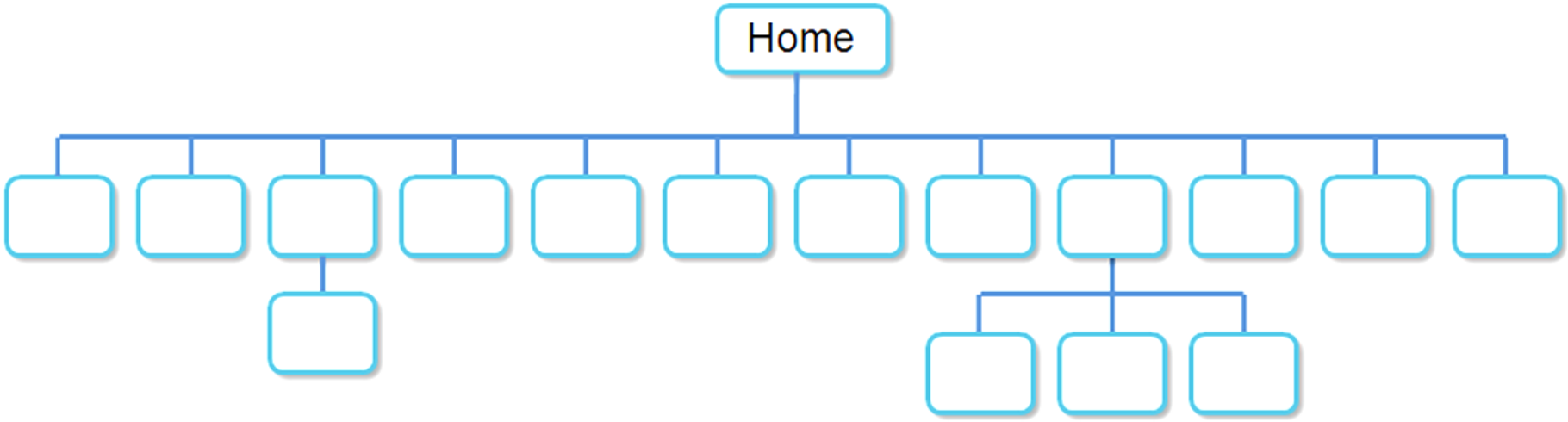
These are the main types of website organization structures.

HIERARCHICAL ORGANIZATION



- A clearly defined home page
- Navigation links to major site sections
- Often used for commercial and corporate websites

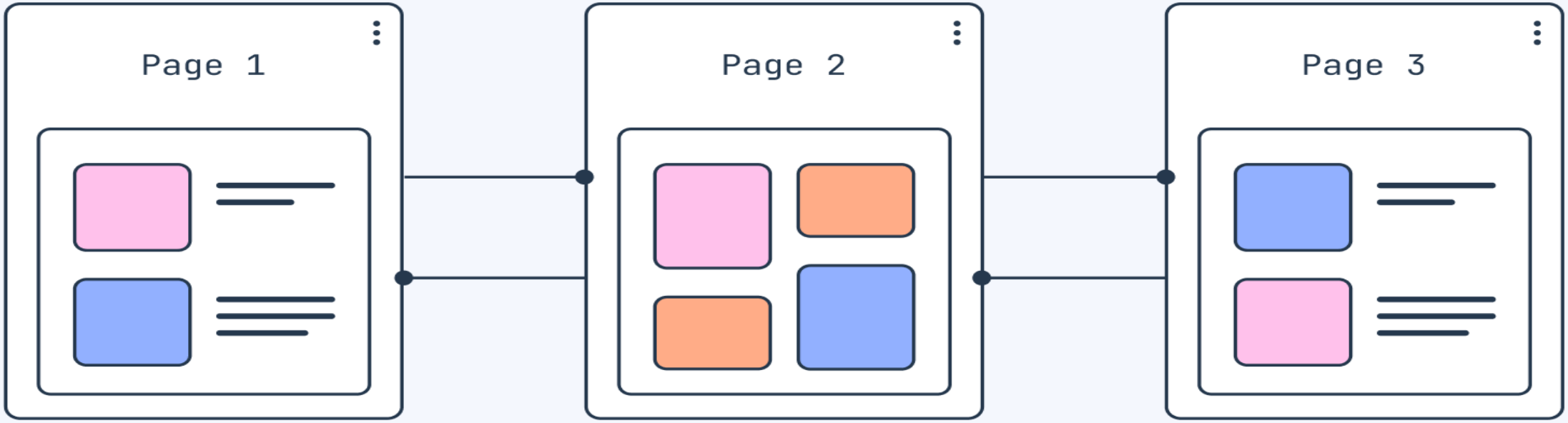
HIERARCHICAL & SHALLOW



BE CAREFUL THAT THE ORGANIZATION IS NOT TOO SHALLOW.

- Too many choices --> a confusing and less usable web site
- Research by Nelson Cowan: adults typically can keep about four items or chunks of items in short-term memory (http://www.ncbi.nlm.nih.gov/pmc/articles/PMC2864034/
- Information Chunking
- Be aware of the number of major navigation links
- Try group navigation links visually into groups with no more than about four links.

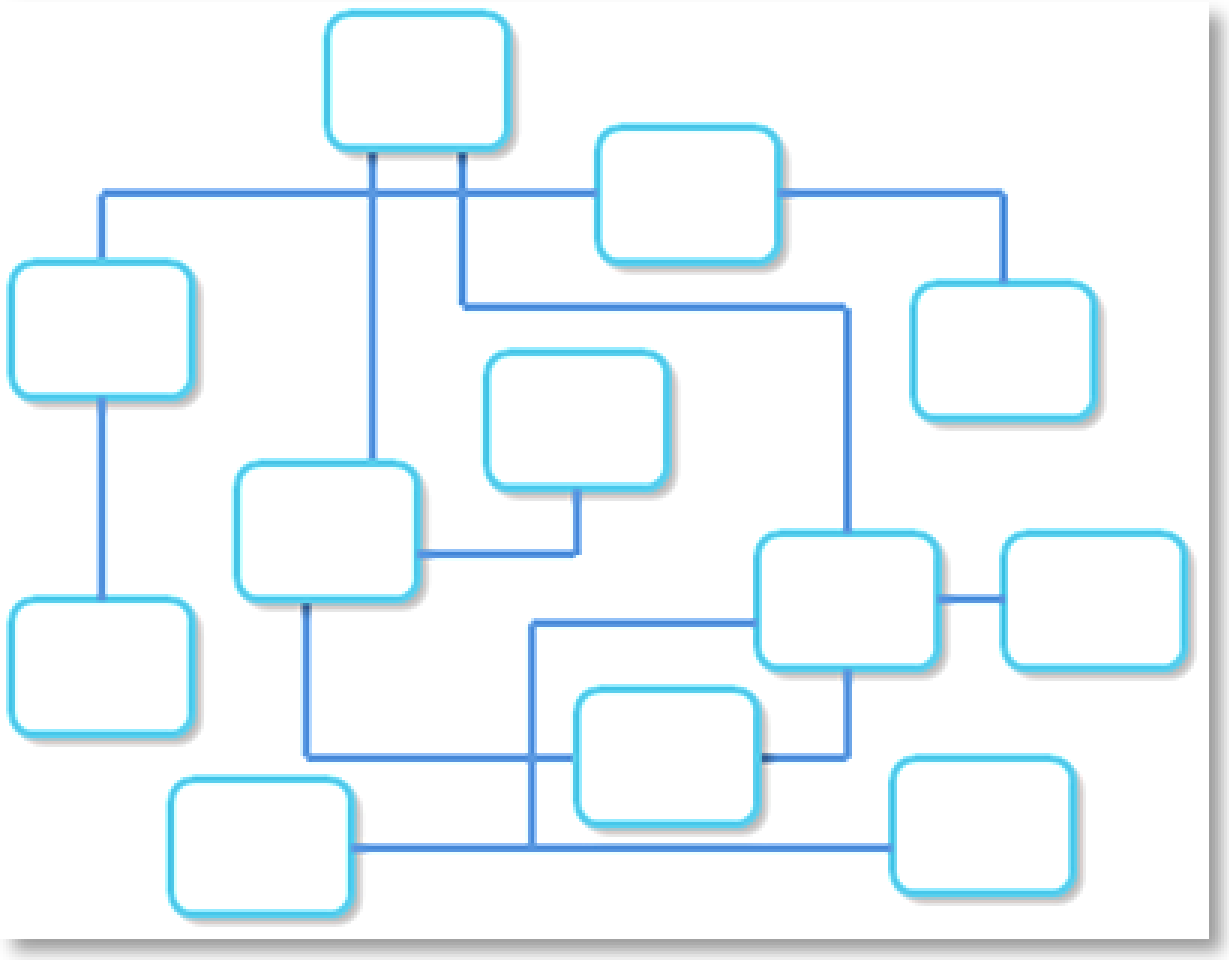
LINEAR ORGANIZATION



slickplan

- A series of pages that provide a tutorial, tour, or presentation.
- Sequential viewing

RANDOM ORGANIZATION



- Sometimes called “Web” Organization
- Usually there is no clear path through the site
- May be used with artistic or concept sites
- Not typically used for commercial sites.

COMPARISON

Type	Structure	Best Use Case	Pros	Cons
Hierarchical	Tree-like	Large websites with a lot of content	Easy to navigate, structured	Can be deep, requiring many clicks
Linear	Sequential	Tutorials, step-by-step processes	Clear direction, easy to follow	Rigid, not ideal for exploration
Random	Web-like	Blogs, creative sites, wikis	Flexible, encourages exploration	Confusing without proper links

DESIGN PRINCIPLES



Repetition

Repeat visual elements like colors, fonts, and shapes throughout the design for consistency.



Contrast

Use contrasting colors, sizes, and styles to add visual interest and draw attention to important elements.



Proximity

Group related content together through alignment, spacing, borders, and sections.



Alignment

Align page elements like text, images, and sections to create visual unity.

Using design principles like repetition, contrast, proximity, and alignment creates organized, visually appealing web page layouts.

DESIGN TO PROVIDE FOR ACCESSIBILITY



Include multiple ways to access content

Using text, images, videos, audio allows people with disabilities to still understand content



Design for different devices

Optimizing for mobile allows those without computers to still access content



Meet legal requirements

Following standards like WCAG 2.0 ensures accessibility and avoids legal issues

Designing for accessibility improves experience for all users and avoids legal problems

DESIGN FOR ACCESSIBILITY

Web Content Accessibility Guidelines 2.0 (WCAG 2.0)

http://www.w3.org/TR/WCAG20/Overview

http://www.w3.org/WAI/WCAG20/quickref

Based on Four Principles (POUR)



Perceivable

Content must be Perceivable



Understandable

Content and controls must be Understandable



Operable

Interface components in the content must be Operable



Robust

Content should be Robust enough to work with current and future user agents, including assistive technologies

Designing for accessibility improves experience for all users and avoids legal problems

WRITING FOR THE WEB



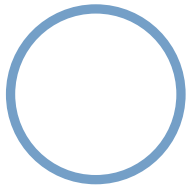
Use headings and subheadings

Break up blocks of text using headings and subheadings to improve readability



Keep paragraphs short

Paragraphs should be no more than 3-4 sentences to improve readability



Use bullet points

Bullet points make content more scannable for readers

By following web writing best practices like using headings, short paragraphs, and bullet points, you can improve the readability and scannability of content for your readers.

He's not a word hero. He's a word hero because he was captured. I like text that wasn't captured. I think the only card she has is the Lorem card. He's not a word hero. He's a word hero because he was captured. I like text that wasn't captured. I'm speaking with myself, number one, because I have a very good brain and I've said a lot of things. When other websites give you text, they're not sending the best. They're not sending you, they're sending words that have lots of problems and they're bringing those problems with us. They're bringing mistakes. They're bringing misspellings. They're typists... And some, I assume, are good words.

Not easy to read

Easier to read

Headers: give a clear structure to your text (and also help with SEO).

Who is he?

He's **not a word hero**. He's a word hero because he was captured. I like text that wasn't captured. I think the only card she has is the Lorem card. He's not a word hero. He's a word hero because he was captured. I like text that wasn't captured.

Fonts matter: You want the text to be legible and pleasant to look at. On all devices and screen sizes!

Look at the negative space: or whitespace on the page. This helps avoid huge walls of text and long sentences. Break it down into smaller sections whenever possible.

Word issues

I'm speaking with myself, number one, because **I have a very good brain** and I've said a lot of things. When other websites give you text, they're not sending the best. They're not sending you, they're sending words that have lots of problems and they're bringing those problems with us:

Go bold: helps highlight the important information for skimmers.

Bulleted lists: such as this one. They help break down the information on the page into digestible chunks.

- They're bringing mistakes.
- They're bringing misspellings.

They're typists... And some, **I assume, are good words**.

Extra media: add images, graphics, and even videos to break down the text.

DESIGN “EASY TO READ” TEXT



Use common fonts

Arial, Helvetica, Verdana, Times New Roman are good options

https://premiumcoding.com/10-most-popular-fonts-in-web-design/



Use appropriate text size

Medium, 1em, 100% are good defaults

https://www.tooltester.com/en/blog/web-typography-best-practices/



Use strong contrast

Between text and background colors

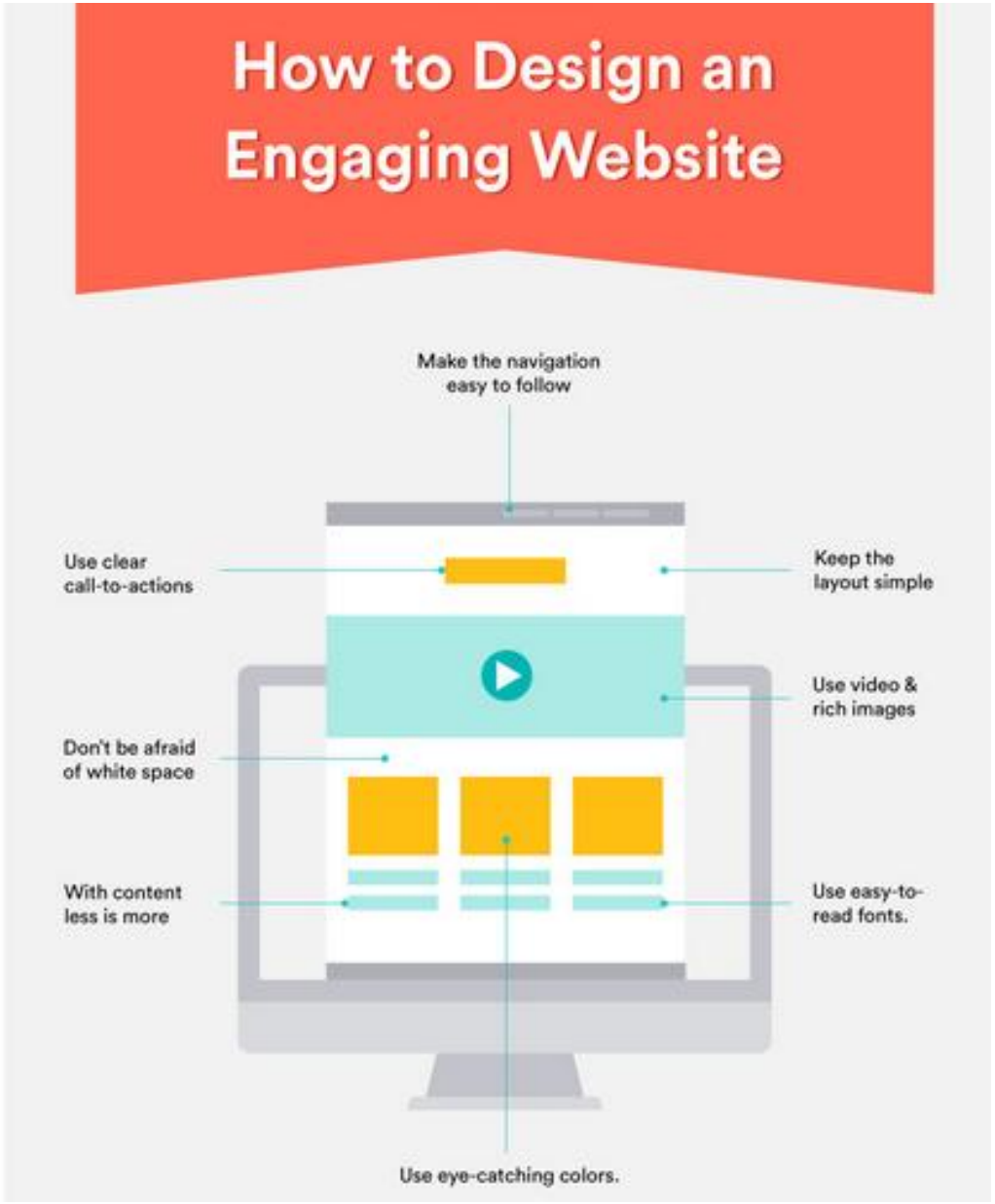
https://cultureconnectme.com/color/



Use columns

Instead of wide horizontal text

Following these tips will create easy to read text



GAIN TRUST WITH A CLEAN, ENGAGING WEBSITE DESIGN

MORE TEXT DESIGN CONSIDERATIONS



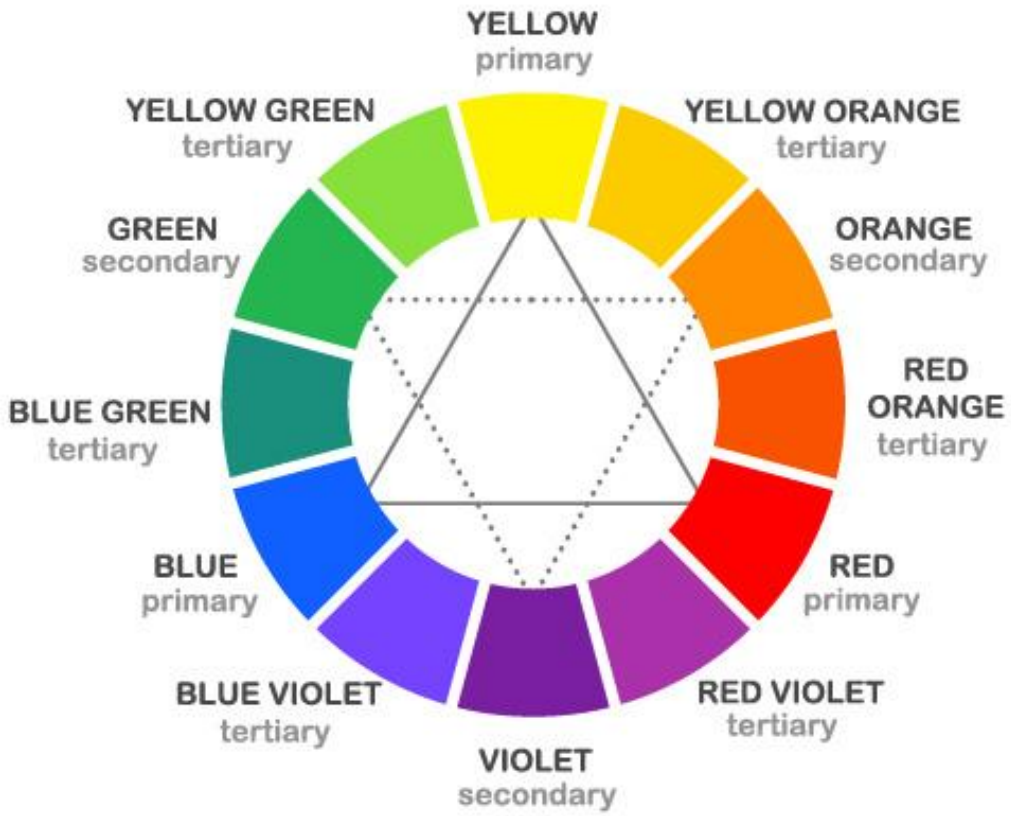
Carefully choose text in hyperlinks

- Avoid “click here”
- Hyperlink key words or phrases, not entire sentences



Chek yur spellin (Check your spelling)

USE COLOR THEORY





PRIMARY

YELLOW
RED
BLUE



SECONDARY

ORANGE
VIOLET
GREEN



TERTIARY

YELLOW ORANGE
RED ORANGE
RED VIOLET
BLUE VIOLET
BLUE GREEN
YELLOW GREEN

COLOR SCHEMES BASED ON THE COLOR WHEEL (1)



Monochromatic – shades, tints, or tones of the same color



Analogous – a main color and two colors adjacent to it on the color wheel



Complementary – two colors that are opposite each other on the color wheel

http://meyerweb.com/eric/tools/color-blend

COLOR SCHEMES BASED ON THE COLOR WHEEL (2)



Split Complementary – a main color, the color opposite it on the color wheel (the complement) and two colors adjacent to the complement



Triadic- three colors that are equidistant on the color wheel



Tetradic – two complementary color pairs

IMPLEMENTING A COLOR SCHEME



Choose a dominant color

Select one main color that will be used most prominently throughout the design



Use accent colors for highlights

Use other colors from the palette for **headings**, **borders**, etc. to create visual interest

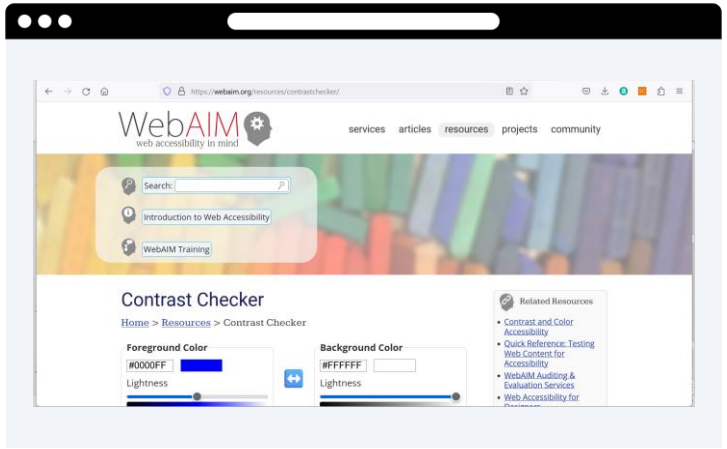


Include neutrals

Use **whites, grays, blacks** to give the eyes a rest and make the colors pop

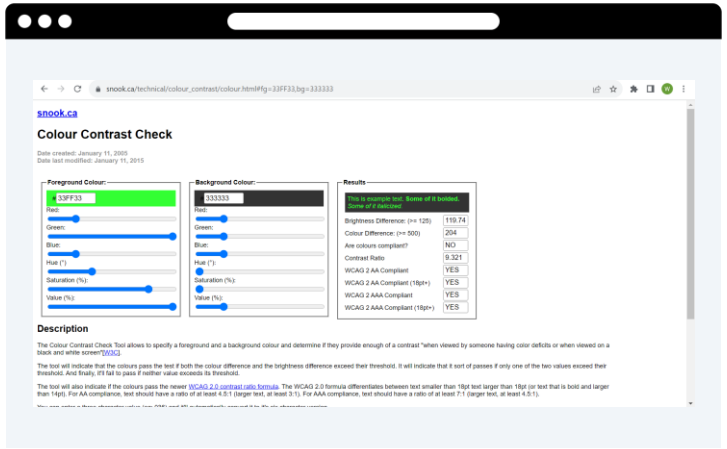
Following these tips will create a cohesive and aesthetically pleasing color scheme.

VERIFY SUFFICIENT CONTRAST



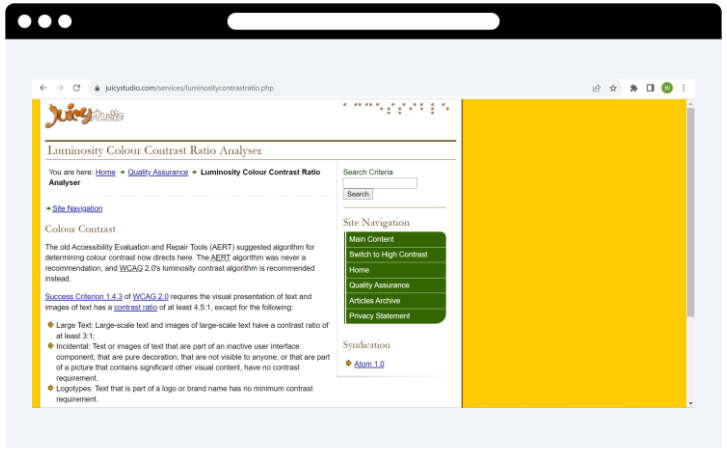
Contrast Checker

https://webaim.org/resources/contrastchecker/



Colour Contrast Check

https://snook.ca/technical/colour_contrast/colour.html#fg=33FF33,bg=333333



Luminosity Colour Contrast Ratio Analyser

https://juicystudio.com/services/luminositycontrastratio.php

Check color contrast to ensure text is readable

COLOR SCHEME RESOURCES



Color scheme resources help in choosing colors

There are many websites that provide tools to generate, visualize and choose color schemes and palettes for web design.



Adobe Kuler is a popular resource

Adobe Kuler is a web-based application that allows users to generate and save color themes and browse thousands of themes created by the community.

- http://kuler.adobe.com



Other tools

Websites like ColorSchemeDesigner.com allow creating color schemes by setting primary colors and getting secondary colors.

- http://meyerweb.com/eric/tools/color-blend
- http://colorschemedesigner.com
- http://www.colorsontheweb.com/colorwizard.asp
- http://www.leestreet.com/QuickColor.swf
- http://www.colorspire.com
- http://colrd.com

There are many great online resources and tools available for choosing color schemes and palettes for web design projects.

COLOR & TARGET AUDIENCES

Target Younger Audiences

Use the color red to capture attention and convey urgency.

Target Construction Industry

Use orange to promote creativity and safety.

Target Natural Foods Industry

Use the color green to align with natural products.

Target Financial Institutions

Use blue to convey trustworthiness.

Target Beauty Industry

Use purple to appeal to female audiences.

CHECKPOINT

List the four basic principles of design.

View the home page of your school and describe how each principle is applied.

Choose any website

Describe the target audience for the site.

Does the site meet the needs of its target audiences?

USE OF GRAPHICS & MULTIMEDIA

- File size and dimension matter

  Keep file sizes small and limit image dimensions for quick loading.

- Provide for robust navigation

  Use intuitive and accessible navigation like breadcrumbs and sitemaps.

- Antialiased/aliased text considerations

  Choose antialiased text for smooth fonts on screens.

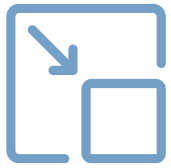
- Provide alternate text

  Use alt text for non-text elements like images for accessibility.

- Use only necessary multimedia

  Limit use of multimedia to critical elements to avoid overloading.

GRAPHIC DESIGN BEST PRACTICES(1)



Limit large graphics

Stick to the 60K size recommendation for graphics



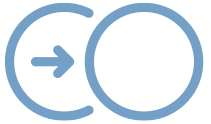
Use alt text

Provide descriptive alternate text for all images



Ensure accessibility

Message remains clear even if images don't load. Provide plain text links.



Guide to use animation

Use animation only if it makes the page more effective and provide a text description

Follow best practices for using graphics and images on web pages.

GRAPHIC DESIGN BEST PRACTICES(2)

There is no requirement to limit your color choices to web safe colors. However, choose colors on the web palette if consistency across older Windows/Mac platforms is important.



Use anti-aliased text in images

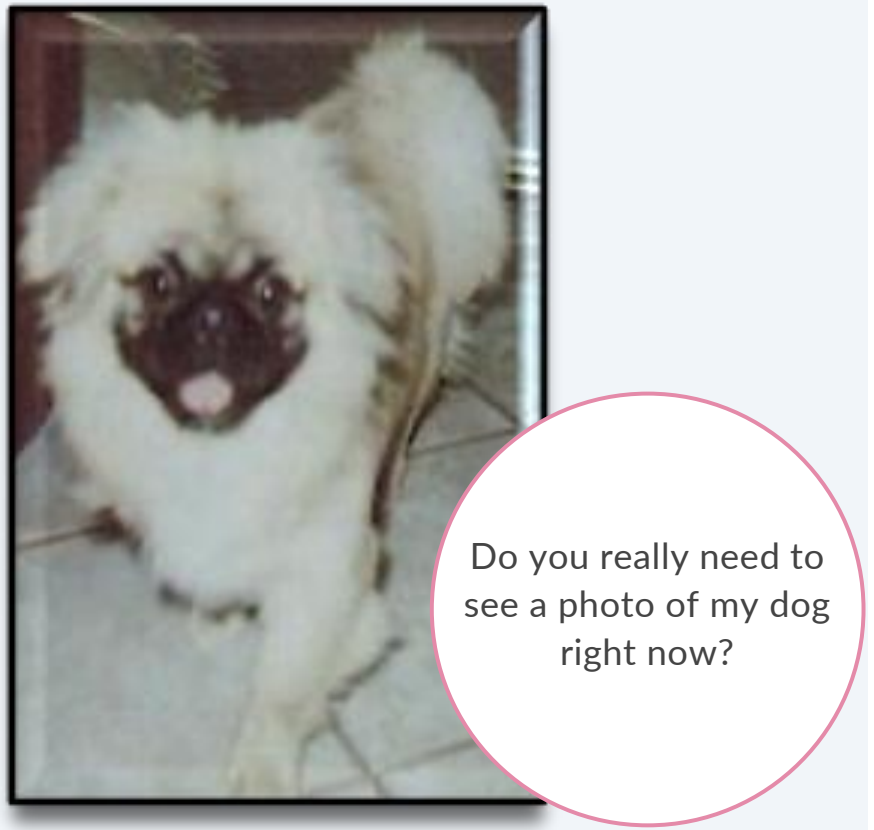


https://fontsaga.com/how-antialiased-font-can-improve-user-experience/

GRAPHIC DESIGN BEST PRACTICES(3)

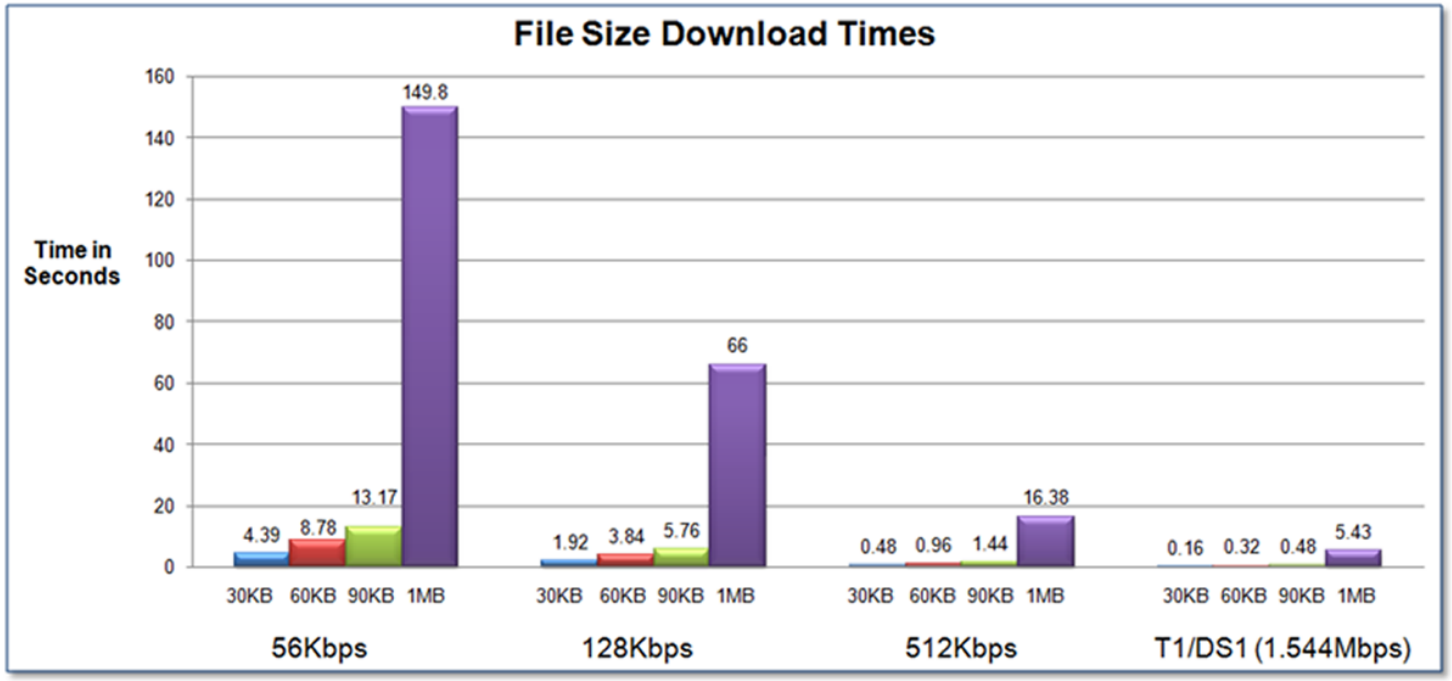
Use only necessary images

Goal: image file size should be as small as possible with acceptable display quality

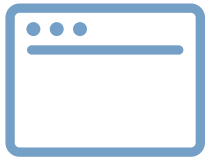


GRAPHIC DESIGN BEST PRACTICES(4)

- Watch the load time of your pages
- Try to limit web page document and associated media to under 60K on the home page

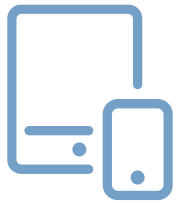


WEB PAGE DESIGN: BROWSERS & SCREEN RESOLUTION



Test in multiple browsers

Test website in Internet Explorer, Chrome, Firefox and Safari.



Test at various resolutions

Test at common resolutions like 1366x768, 1024x768, 1280x800, 1280x1024.



Design for different resolutions

Use centered content and fixed/responsive width to support various resolutions.

Testing across browsers and resolutions ensures a website works for all users.

NAVIGATION DESIGN



Make navigation easy to find

Place main navigation consistently across all pages, like top or left side



Consider different navigation options

Navigation bars, breadcrumbs, graphics, dynamic menus, site map, search, skip links

Good navigation design improves usability and accessibility.

WIREFRAME

Logo

The logo sits at the top of the page and represents the brand.

Navigation

The navigation menu allows users to navigate between different pages on the website.

Content

The main content of the page goes in the center and contains the information and media that users came to the site for.

Footer

The footer at the bottom often contains secondary information like contact details, sitemap, privacy policy etc.

WIREFRAME

A SKETCH OF BLUEPRINT OF A WEB PAGE

Shows the structure of the basic page elements, including:

- Logo
- Navigation
- Content
- Footer



WEB PAGE DESIGN PAGE LAYOUT (1)

- Place the most important information "above the fold"
- Use adequate "white" or blank space
- Use an interesting page layout

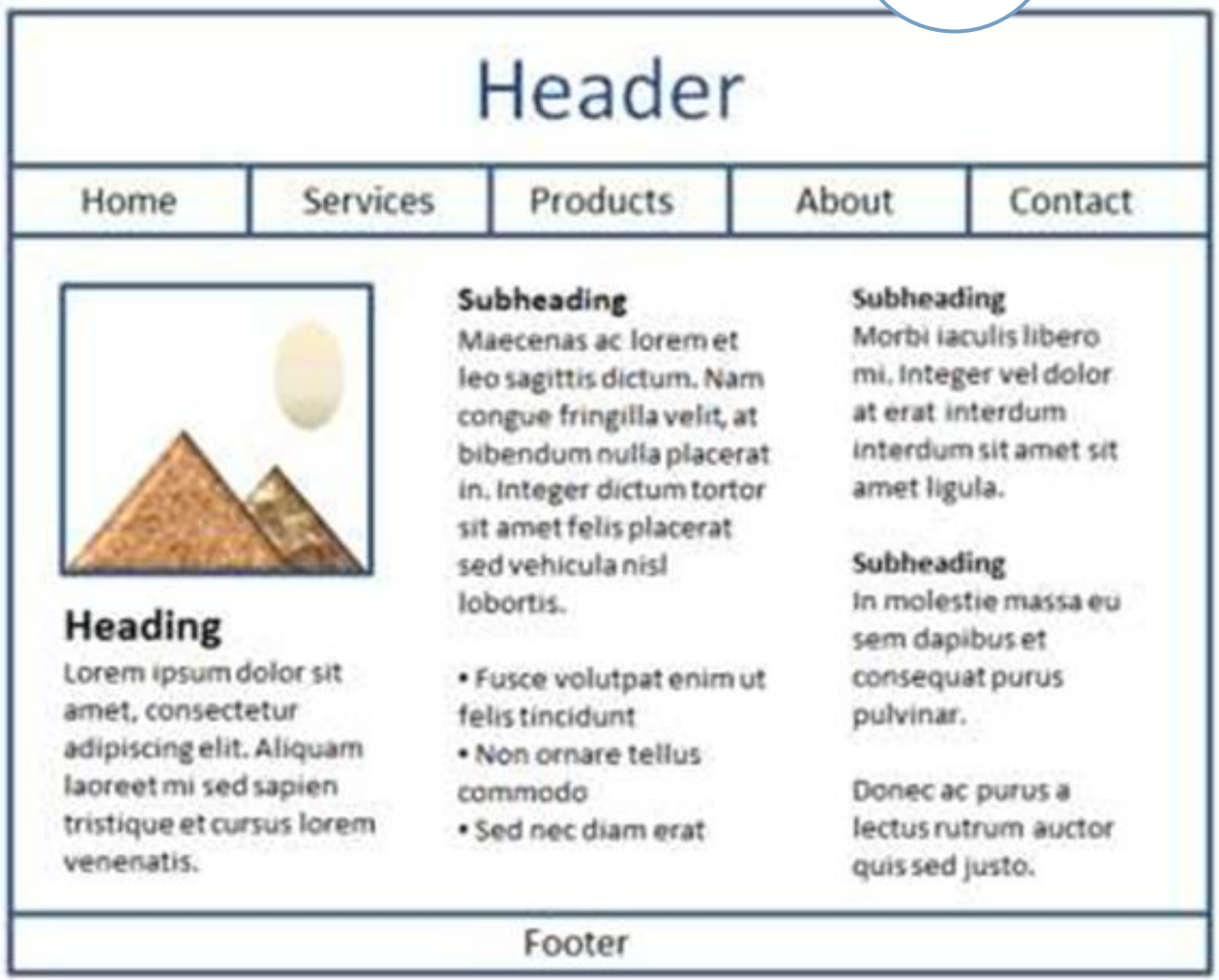


This is usable, but a little boring. See the next slide for improvements in page layout.

WEB PAGE DESIGN PAGE LAYOUT (2)

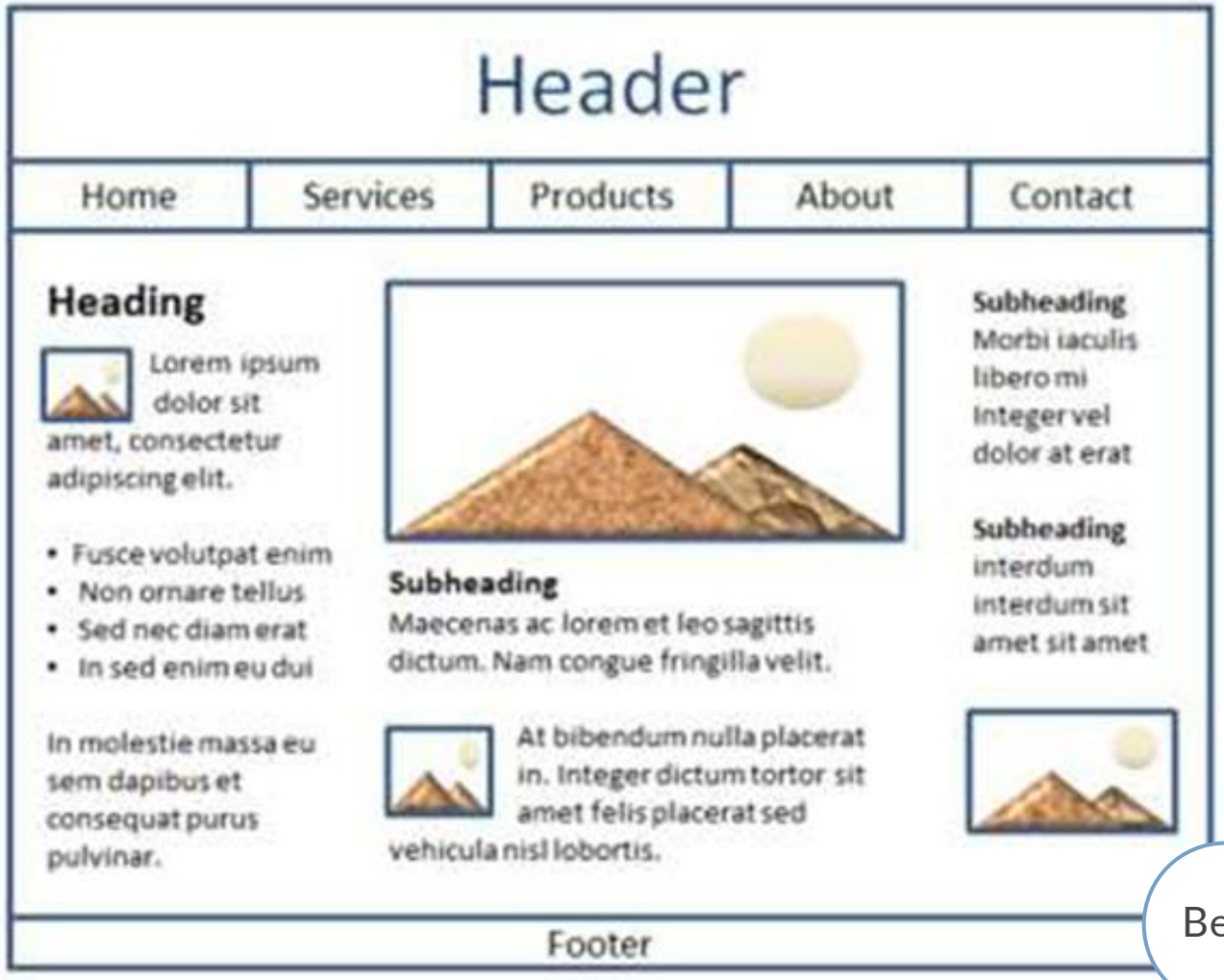
- Columns make the page more interesting and it's easier to read this way.

Better

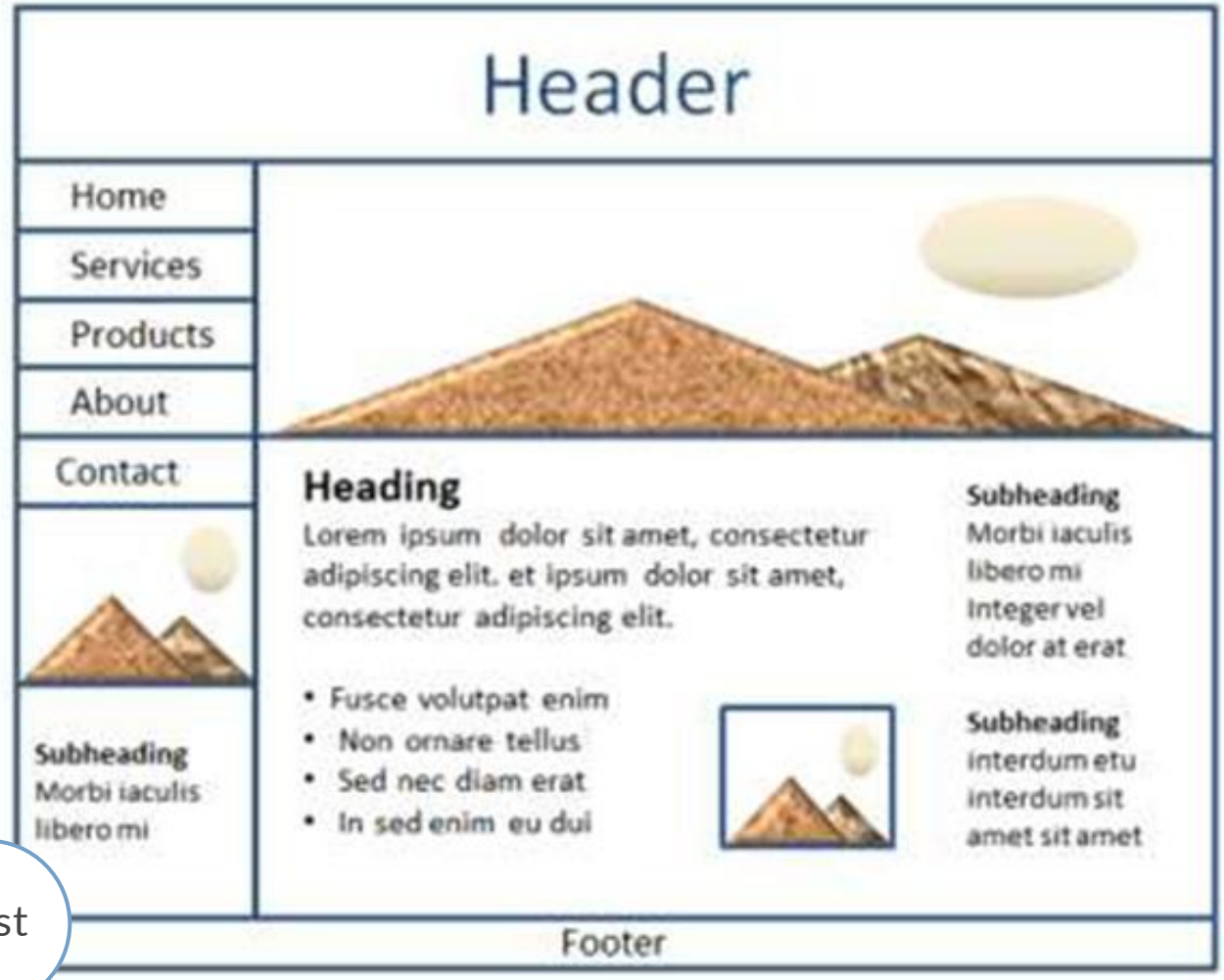


WEB PAGE DESIGN PAGE LAYOUT (3)

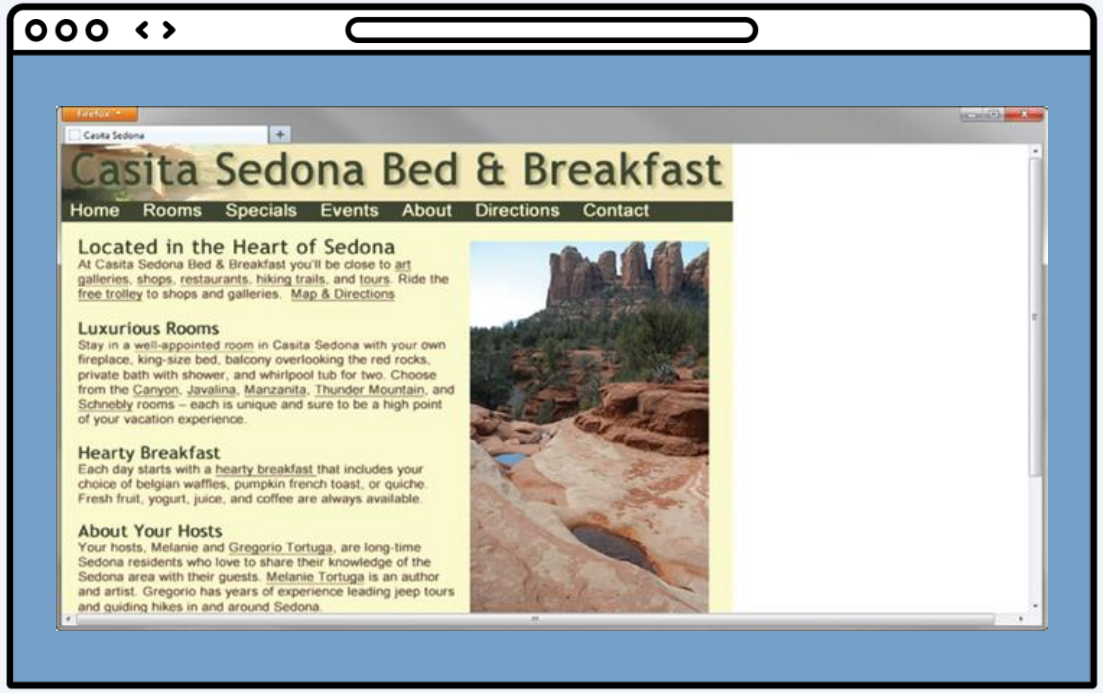
Columns of different widths interspersed with graphics and headings create the most interesting, easy to read page



Best



PAGE LAYOUT DESIGN TECHNIQUES



Fixed Layout

AKA rigid or “ice” design

Fixed-width often at left margin



More appealing if fixed with content is centered

PAGE LAYOUT DESIGN TECHNIQUES





Fluid Layout

AKA liquid design

Expands to fill the browser at all resolutions.

Adaptation:

Page content typically centered and
often configured with a percentage width such as 80%

DESIGN FOR THE MOBILE WEB



Mobile usage growing

By 2015 more people will access the web via mobile than desktop. Keyword: mobile phone



Three approaches

Separate mobile site, mobile subdomain, or responsive design. Keyword: options

With mobile usage rising, web design must be optimized for smaller screens.

MOBILE DESIGN QUICK CHECKLIST

- **Small screen size**
  Optimize layout for small screens like smartphones
- **Bandwidth issues**
  Minimize large files like images and videos
- **Single-column layout**
  Use a single column layout for better readability
- **Maximize contrast**
  Use high contrast colors for better visibility
- **Optimize images**
  Compress and resize images for faster loading
- **Descriptive alt text**
  Provide descriptive alternate text for screen readers
- **Avoid non-essential content**
  Remove unnecessary content to minimize bandwidth

RESPONSIVE WEB DESIGN

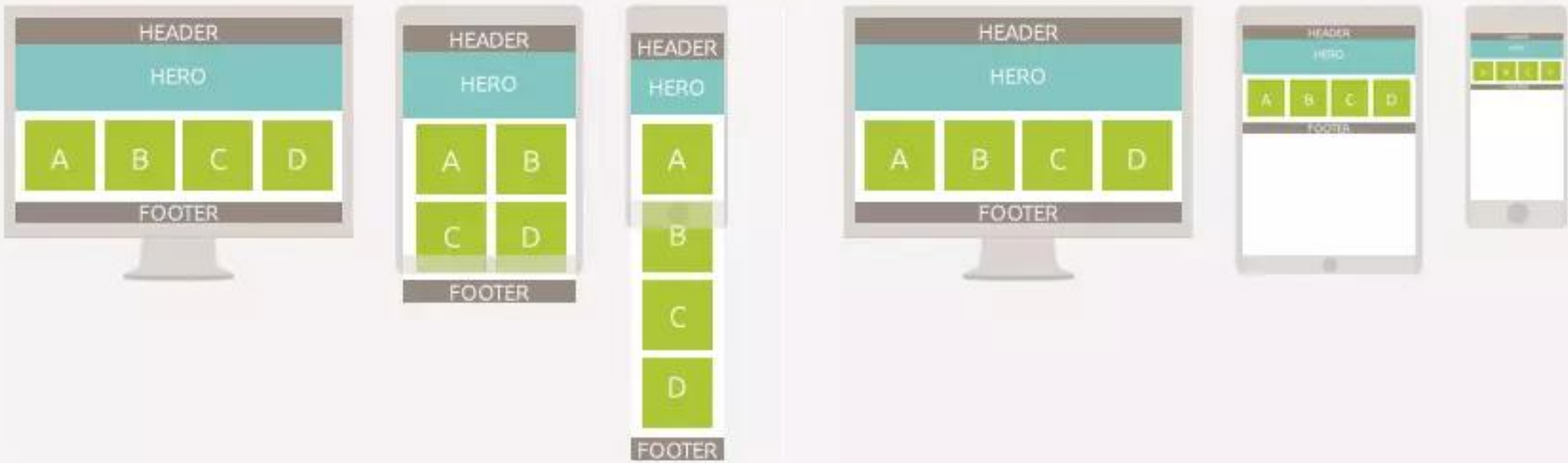


Ethan Marcotte, noted web developer
http://www.alistapart.com/articles/responsive-web-design

Progressively enhancing a web page for different viewing contexts (such as smartphones and tablets) through the use of coding techniques, including flexible layouts and media queries.

Examples: http://www.mediaqurie.es

RESPONSIVE VS NON RESPONSIVE



Responsive Web Design
Text, images and links easy to read on all devices

Standard Web Design (non-responsive)
Text, images and links difficult to read on mobile devices

WEB DESIGN BEST PRACTICES CHECKLIST

- **Page layout**
  Use grid-based layouts with CSS for flexible resizing across devices.
- **Browser compatibility**
  Test site on major browsers like Chrome, Firefox, Safari, etc. and ensure core functionality works on all.
- **Navigation**
  Use simple, intuitive navigation like hamburger menus for mobile and clear tabs/links on desktop.
- **Color and graphics**
  Use a consistent color scheme and appropriate image formats like SVG for best performance.
- **Multimedia**
  Ensure videos and audio play cross-browser and don't autoplay to avoid disrupting users.
- **Content presentation**
  Present content clearly with good typography and spacing for easy scanning.
- **Functionality**
  Site features should work reliably cross-browser and on all devices.
- **Accessibility**
  Follow standards like ARIA roles to support screen readers and keyboards.

CHECKPOINT

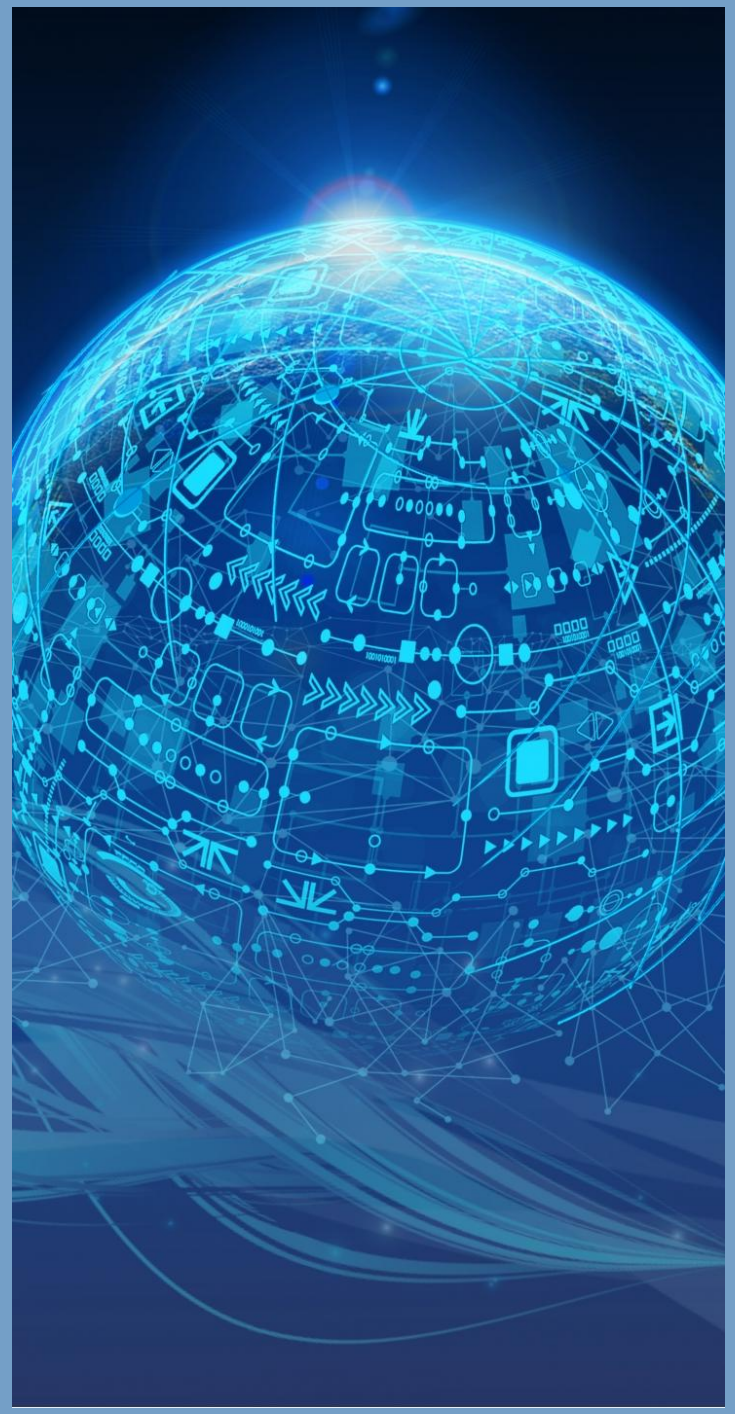
Best Practices

View the home page of your school. Use the Best Practices Checklist (Table 5.1) to evaluate the page. Describe the results.

View your favorite web site

Maximize and resize the browser window.

Decide whether the site uses fixed or fluid design.
Adjust the screen resolution on your monitor
(Start > Control Panel > Display > Settings) to a different resolution than you normally use.
Does the site look similar or very different?
List two recommendations for improving the design of the site.



- This chapter introduced you to best practices of web design.
- The choices you make in the use of color, graphics, and text should be based on your particular target audience.